Drought Severity Mapping using Vegetation Condition Index (VCI)

Project Report Submitted in fulfillment of the Requirements for the Award of the Internship of Summer Training Program Space Science Technology

Remote Sensing and GIS using Python

By
P Sajai Krishna
NSS College of Engineering
NSS24EC109
913269

Under the Supervision of
Miss. Alisha Sinha



India Space Academy,
Department of Space Education, India Space Week

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TITLE

Drought Severity Assessment using Multi-temporal Vegetation

Condition Index from Satellite Data

OBJECTIVE

To assess agricultural or regional drought severity by analyzing vegetation condition using multi-temporal satellite data and deriving the Vegetation Condition Index (VCI), enabling identification of drought-affected zones and vegetation stress patterns.

STUDY AREA

The study area lies in Rajkot district, Gujarat, India, in the semi-arid Saurashtra region. The selected Area of Interest (AOI) spans the area between Rajkot, Gondal, Shapar, and Jasdan.

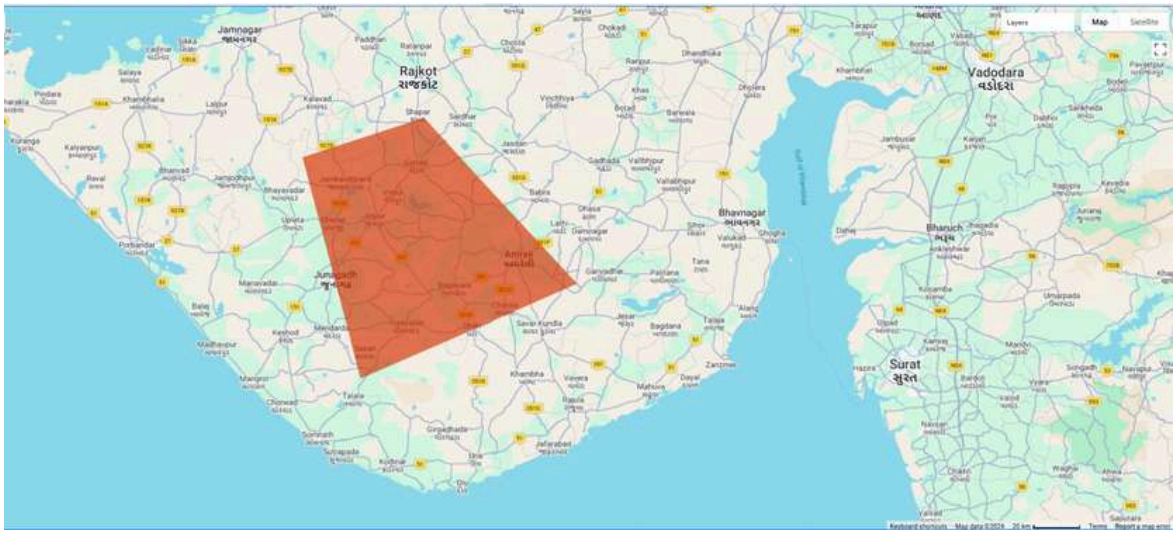


Figure 1: Study Area Map

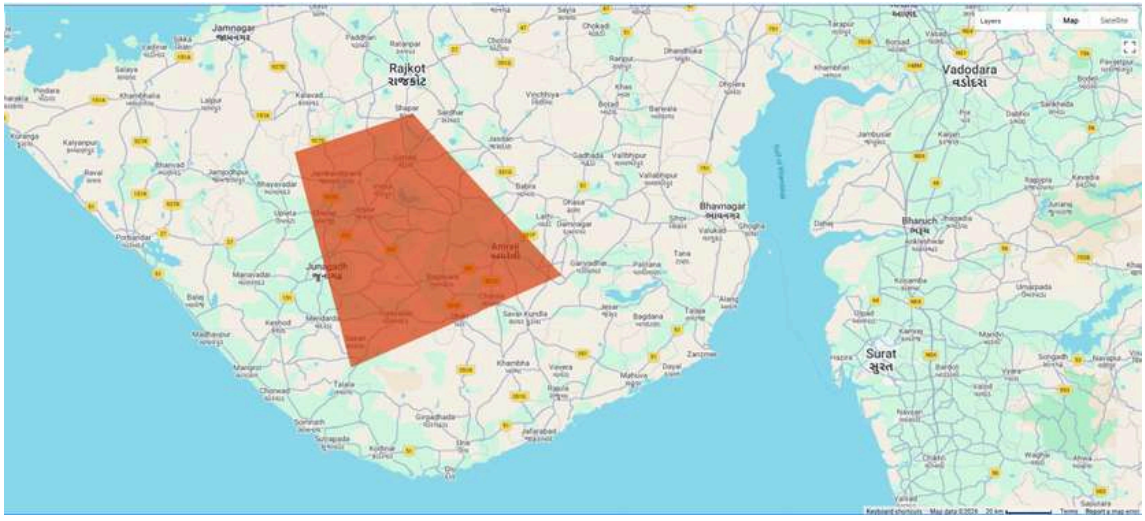
DATA USED

Satellite Data

MODIS NDVI (MOD13Q1 Product) — used for time-series drought analysis

Auxillary Data

The Area of Interest (AOI) was manually delineated using the geometry drawing tool available in the Google Earth Engine Code Editor. The AOI is the region within Rajkot district, Gujarat.



Data Source

The MODIS NDVI dataset was accessed through the Google Earth Engine Data Catalog: https://developers.google.com/earth-engine/datasets/catalog/MODIS_006_MOD13Q1

Date Range:

- Historical baseline: January 1, 2015 – December 31, 2023
- Current assessment period: January 1, 2023 – December 31, 2023

Reference / Validation Data

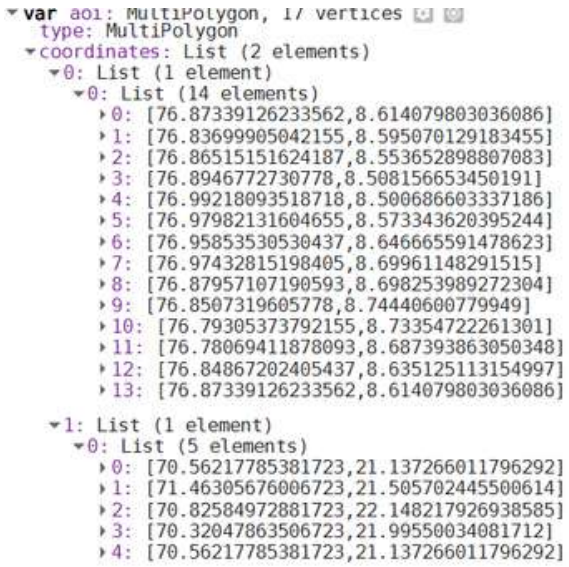
These factors were not considered in this study.Cross-validation was not performed against Google Earth Pro, rainfall datasets or existing drought reports.

METHODOLOGY

This project was done using the Google Earth Engine (GEE) Code Editor. The methodology followed for drought severity assessment using the Vegetation Condition Index (VCI) is described below.

1. Defining the Area of Interest (AOI)

The Area of Interest was defined manually using the geometry drawing tool in the Google Earth Engine Code Editor. A polygon was drawn over the Rajkot district region in Gujarat, India,



2. Loading the NDVI Image Collection

The MODIS MOD13Q1 dataset was loaded into Google Earth Engine and filtered according to the study area and the analysis period from January 1, 2015 to December 31, 2023.

```
// Load MODIS NDVI time series
var ndvi = ee.ImageCollection("MODIS/006/MOD13Q1")
  .filterBounds(aoi)
  .filterDate('2015-01-01', '2023-12-31')
  .select('NDVI');
```

3. Compute NDVI minimum and maximum:

The minimum and maximum NDVI for the years 2015 to 2023 were calculated for every pixel in the study area. These two parameters represent the historical status of vegetation and form the baseline for VCI calculation.

```
// Min and max NDVI over the whole period (per pixel)
var ndvi_min = ndvi.min();
var ndvi_max = ndvi.max();
```

4. Select a target year or season NDVI:

The mean NDVI of the year 2023 was used to identify the current vegetation status in the study area and compared with the baseline NDVI, which was calculated for the years 2015 to 2023.

```
// Current period to assess (change dates if you want a different year)
var ndvi_current = ndvi.filterDate('2023-01-01','2023-12-31').mean();
```

5. Calculating the Vegetation Condition Index (VCI)

Vegetation Condition Index (VCI) is calculated using the following formula:

$$VCI = \frac{NDVI_{current} - NDVI_{min}}{NDVI_{max} - NDVI_{min}}$$

The VCI indicates the relative condition of vegetation compared to its historical range.

```
// Calculate VCI
var vci = ndvi_current.subtract(ndvi_min)
  .divide(ndvi_max.subtract(ndvi_min))
  .rename('VCI');
```

6. Visualize drought severity:

Visualization of the resulting image of the Vegetation Condition Index was done using the red-yellow-green color palette, where red corresponds to severe vegetation stress, yellow represents moderate stress, and green represents healthy vegetation conditions.

```
// Center map and show the result
Map.centerObject(aoi, 9);
Map.addLayer(vci, {min:0, max:1, palette:['red','yellow','green']}, 'VCI');
```

7. Classify drought zones using thresholds and visualize results.

The VCI values were classified into four drought severity categories:

```
// Drought class breakdown
var extreme = vci.lt(0.2);
var moderate = vci.gte(0.2).and(vci.lt(0.4));
var mild = vci.gte(0.4).and(vci.lt(0.6));
var normal = vci.gte(0.6);

var pixelArea = ee.Image.pixelArea().divide(1e6);

function classArea(mask) {
  return pixelArea.updateMask(mask).reduceRegion({
    reducer: ee.Reducer.sum(),
    geometry: aoi,
    scale: 250,
    maxPixels: 1e13
  });
}

print('Extreme drought area (sq km):', classArea(extreme));
print('Moderate drought area (sq km):', classArea(moderate));
print('Mild drought area (sq km):', classArea(mild));
print('Normal area (sq km):', classArea(normal));
```

- VCI < 0.2 : Extreme Drought
- 0.2 ≤ VCI<0.4: Moderate Drought
- 0.4 ≤ VCI<0.6: Mild Drought
- VCI ≥ 0.6: Normal/Healthy Vegetation

8. Generate statistics using reducers:

The area of the drought classes was calculated based on masking the VCI map for each drought class and calculating the total area of the pixels within the AOI using the reduceRegion() function in Google Earth EngineThe results were expressed in square kilometers.

```
// Print mean VCI value for AOI
var stats = vci.reduceRegion({
  reducer: ee.Reducer.mean(),
  geometry: aoi,
  scale: 250,
  maxPixels: 1e13
});
print('Mean VCI for AOI:', stats);
```

9. Export drought maps:

```
// Export VCI map
Export.image.toDrive({
  image: vci,
  description: 'VCI_Drought_Map',
  scale: 250,
  region: aoi
});

// Export classified drought map
Export.image.toDrive({
  image: droughtClassified,
  description: 'Drought_Classified_Map',
  scale: 250,
  region: aoi
});
```

CODE:

```
Imports (1 entry)
var aoi: MultiPolygon, 17 vertices
  type: MultiPolygon
  coordinates: List (2 elements)
    0: List (1 element)
      0: List (14 elements)
    1: List (1 element)
      0: List (5 elements)
        0: [70.56217785381723,21.137266011796292]
        1: [71.46305676006723,21.505702445500614]
        2: [70.82584972881723,22.148217926938585]
        3: [70.32047863506723,21.99550034081712]
        4: [70.56217785381723,21.137266011796292]
```

```
// Load MODIS NDVI time series
var ndvi = ee.ImageCollection("MODIS/006/MOD13Q1")
  .filterBounds(aoi)
  .filterDate('2015-01-01', '2023-12-31')
  .select('NDVI');

// Min and max NDVI over the whole period (per pixel)
var ndvi_min = ndvi.min();
var ndvi_max = ndvi.max();

// Current period to assess (change dates if you want a different year)
var ndvi_current = ndvi.filterDate('2023-01-01','2023-12-31').mean();

// Calculate VCI
var vci = ndvi_current.subtract(ndvi_min)
  .divide(ndvi_max.subtract(ndvi_min))
  .rename('VCI');

// Center map and show the result
Map.centerObject(aoi, 9);
Map.addLayer(vci, {min:0, max:1, palette:['red','yellow','green']}, 'VCI');

// Print mean VCI value for your AOI
var stats = vci.reduceRegion({
  reducer: ee.Reducer.mean(),
  geometry: aoi,
  scale: 250,
  maxPixels: 1e13
});
print('Mean VCI for AOI:', stats);
```

```
// Drought class breakdown (extreme/moderate/mild/normal)
var extreme = vci.lt(0.2);
var moderate = vci.gte(0.2).and(vci.lt(0.4));
var mild = vci.gte(0.4).and(vci.lt(0.6));
var normal = vci.gte(0.6);

var pixelArea = ee.Image.pixelArea().divide(1e6); // sq km per pixel

function classArea(mask) {
  return pixelArea.updateMask(mask).reduceRegion({
    reducer: ee.Reducer.sum(),
    geometry: aoi,
    scale: 250,
    maxPixels: 1e13
  });
}

print('Extreme drought area (sq km):', classArea(extreme));
print('Moderate drought area (sq km):', classArea(moderate));
print('Mild drought area (sq km):', classArea(mild));
print('Normal area (sq km):', classArea(normal));
Export.image.toDrive({
  image: vci,
  description: 'VCI_Drought_Map',
  scale: 250,
  region: aoi
});
```

RESULTS

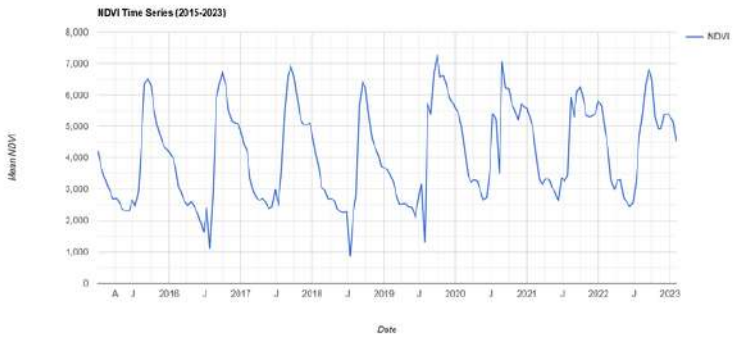
1. VCI Map

The classified VCI map (Figure 2) shows the spatial distribution of vegetation condition across the AOI, with red indicating drought-stressed areas, yellow indicating moderate stress, and green indicating healthy vegetation.



Figure 2: VCI map of the AOI for 2023,

2. NDVI time-series datasets



3. Classified drought zones

Each pixel is classified into one of four drought severity categories based on standard VCI thresholds:

- Red → Moderate drought (VCI 0.2–0.4)
- Orange → Mild drought (VCI 0.4–0.6)
- Yellow → Normal/healthy vegetation (VCI ≥ 0.6)
- Green → Extreme drought (VCI < 0.2)



4. Drought Class Area Statistics

Drought Class	VCI Range	Area (sq. km)
Extreme drought	<0.2	28.55325527140008
Moderate drought	0.2 – 0.4	810.0521491106931
Mild drought	0.4 – 0.6	3197.4233447884317
Normal/healthy	≥ 0.6	3642.9942377835423



Figure 3: Console output showing mean VCI and area statistics by drought class.

Mean VCI:

The mean VCI value for the entire AOI was
found to be 0.581969107166185, indicating moderately healthy vegetation conditions

Conclusion

The Vegetation Condition Index (VCI) provides an effective approach for drought monitoring by normalizing vegetation conditions across time and identifying deviations from normal growth patterns. This method allows accurate iden- tification of drought-affected regions and vegetation stress zones using satel- lite data.

While VCI-based analysis depends on the availability of long-term NDVI datasets and may be influenced by seasonal variability, integrating multi- temporal data significantly improves reliability. This project equips participants with essential skills in drought assessment, time-series analysis, and environmen- tal monitoring, which are highly relevant for agriculture, climate studies, and disaster management applications.

References

1. MODIS NDVI (MOD13Q1 Product) — Google Earth Engine Data Catalog:
   https://developers.google.com/earth- engine/datasets/catalog/MODIS_006_MOD13Q1

2. Google Earth Engine Code Editor:
   https://code.earthengine.google.com

3. India Space Academy, Summer Training Program 2026 — Project Work Guidelines and Methodology Document.